



Problems (Limits and Derivatives)

Limits

Evaluate the following limits

1. $\lim_{x \rightarrow -2} \left(\frac{\frac{1}{x} + \frac{1}{2}}{x+2} \right)$
2. $\lim_{n \rightarrow \infty} (4/n + 5/n)^n$
3. If $f(x) = \begin{cases} |x| + 1, & x < 0 \\ 0, & x = 0 \\ |x| - 1, & x > 0 \end{cases}$, for what values of a does $\lim_{x \rightarrow a} f(x)$ exist?
4. $\lim_{n \rightarrow \infty} (1/n + 2/n + \dots n/n)$
5. $\lim_{x \rightarrow a} \left(\frac{x^x - x^a}{a^x - a^a} \right)$

Derivatives

1. Find the slope of the tangent to the circle $x^2 + y^2 = 13$ at the point (2,3)
2. If $f(x) = \frac{x^{100}}{100} + \frac{x^{99}}{99} + \dots + \frac{x^2}{2} + x + 1$, show that $f'(1) = 100 f'(0)$
3. Assume that $h(x) = f(g(x))$, where both f and g are differentiable functions. If $g(-1)=2$, $g'(-1)=3$, and $f'(2)=-4$, what is the value of $h'(-1)$?
4. Find two nonnegative numbers whose sum is 9 and so that the product of one number and the square of the other number is a maximum.
5. Find the point (x, y) on the graph of $y = \sqrt{x}$ nearest the point (4, 0).
6. Car B is 30 miles directly east of Car A and begins moving west at 90 mph. At the same moment car A begins moving north at 60 mph. What will be the minimum distance between the cars and at what time t does the minimum distance occur?